CS 771A: Intro to Machine Learning, IIT Kanpur				**Quiz I** (29 Jan 2026)
Name	MELBO			**20 marks**
Roll No	260007	Dept.	AWSM	Page **1** of **2**

Instructions:
1. This question paper contains 1 page (2 sides of paper). Please verify.
2. Write your name, roll number, department above in **block letters neatly with ink**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – such cases may get straight 0 marks.
5. Do not rush to fill in answers. You have enough time to solve this quiz.

Q1 (Mix n Match) For each function, select all properties satisfied by that function. A property should be selected only if the function satisfies it for all $x \in \mathbb{R}$. If a function satisfies no property, select NOTA. *Hint: Affine functions look like $f(x) = ax + b$ for some const $a, b \in \mathbb{R}$.* **(5 marks)**

	Function	Continuous?	Differentiable?	Convex?	Affine?	NOTA
a.	$f(x) = \sin(x)$	●	●	○	○	○
b.	$g(x) = \max\{x, 0\} - \max\{-x, 0\}$	●	●	●	●	○
c.	$h(x) = \text{sign}(x), \text{sign}(0) \stackrel{\text{def}}{=} 0$	○	○	○	○	●
d.	$p(x) = \exp(x^2)$	●	●	●	○	○
e.	$q(x) = 1$ (constant function)	●	●	●	●	○

Q2 (Least Squares SVM?) Melba wants to apply least squares to binary classification due to its simplicity. Given N data points $(\mathbf{x}^i, y^i), i = 1, \ldots, N$ with $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \{-1, +1\}$, Melba tries to solve $\min_{\mathbf{w} \in \mathbb{R}^d} \frac{1}{2} \cdot \|\mathbf{w}\|_2^2 + \sum_{i \in [N]} \left(\mathbf{w}^\top \mathbf{x}^i - y^i\right)^2$ (hidden bias). Melbo says that if the labels are either -1 or $+1$, then this is equivalent to solving $\min_{\mathbf{w} \in \mathbb{R}^d} \frac{1}{2} \cdot \|\mathbf{w}\|_2^2 + \sum_{i \in [N]} \ell(z^i)$ where $z^i \stackrel{\text{def}}{=} y^i \cdot \mathbf{w}^\top \mathbf{x}^i$ i.e., ℓ has a form similar to the hinge loss. In the space provided, show that Melbo is right. Write the equation of ℓ, sketch its graph approximately, and show brief derivation. **(2+1+2 = 5 marks)**

Write equation of loss function ℓ here

$$\ell(z) = (1 - z)^2$$

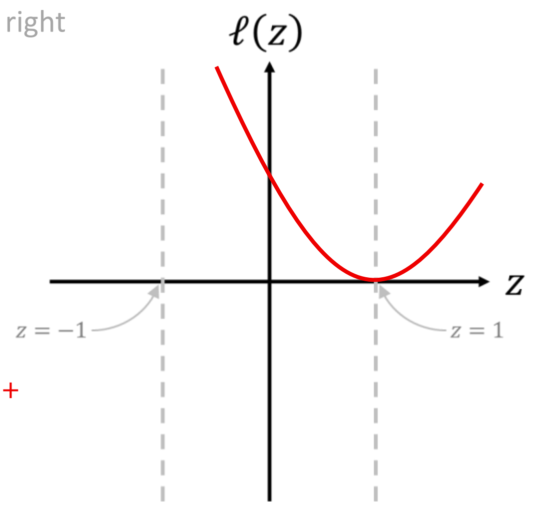
Give brief justification and approximately sketch the graph of ℓ on the right

We have $\left(\mathbf{w}^\top \mathbf{x}^i - y^i\right)^2 = \left(y^i\right)^2 \left(\frac{\mathbf{w}^\top \mathbf{x}^i}{y^i} - 1\right)^2$.

However, if $y^i \in \{-1, +1\}$, then $\frac{1}{y^i} = y^i$ and $\left(y^i\right)^2 = 1$.

Thus, $\left(\mathbf{w}^\top \mathbf{x}^i - y^i\right)^2 = \left(y^i \cdot \mathbf{w}^\top \mathbf{x}^i - 1\right)^2$.

Note that $([1 - z]_+)^2$ is not a correct answer but $[(1 - z)^2]_+$ is a correct answer since $[(1 - z)^2]_+ = (1 - z)^2$.



Q3 (Melbo's Mistake) Melbo wants to simplify the SVM by maximizing the mean distance to the hyperplane and not the smallest distance. For N points $(\mathbf{x}^i, y^i), i \in [N]$, $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \{-1,+1\}$, Melbo wants to solve $\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{|\mathbf{w}^\top \mathbf{x}^i|}{\|\mathbf{w}\|_2}\right\}$ s.t. $y^i \cdot \mathbf{w}^\top x^i \geq 0$ for all i (no bias term). Simplifying as we did in class gives $\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2}\right\}$ s.t. $y^i \cdot \mathbf{w}^\top x^i \geq 0$ for all i. Melbo argues that since the constraints $y^i \cdot \mathbf{w}^\top \mathbf{x}^i \geq 0$ have already been used to simplify, we can remove them and just solve $\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2}\right\}$ without any constraints. Show that Melbo is incorrect. **(6+2+2 = 10 marks)**

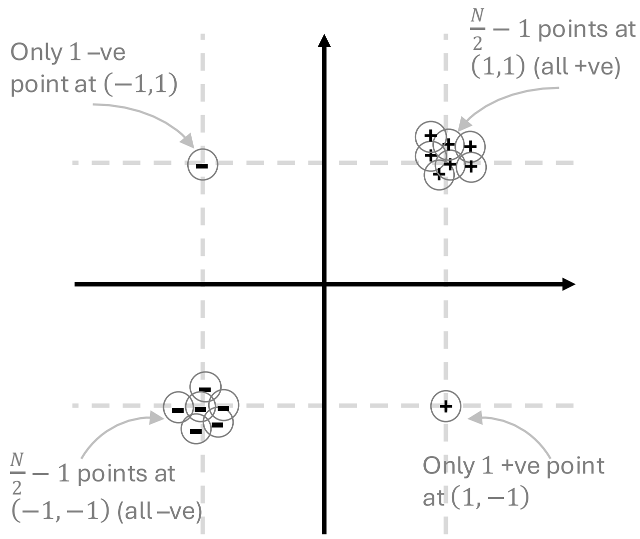
(a) Solve $\arg\max_{\mathbf{w}\in\mathbb{R}^d} \left\{\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2}\right\}$. Assume $\sum_{i\in[N]} y^i \cdot \mathbf{x}^i \neq \mathbf{0}$. Multiple solutions exist (easy to see that if $\mathbf{w}$ is a solution, so is $c \cdot \mathbf{w}$ for $c > 0$). Thus, please give a unit norm solution i.e. $\|\mathbf{w}\|_2 = 1$.

Give brief derivation for the unit norm solution to the problem

(a) We note that by linearity, $\sum_{i\in[N]} \frac{y^i \cdot \mathbf{w}^\top \mathbf{x}^i}{\|\mathbf{w}\|_2} = \hat{\mathbf{w}}^\top \boldsymbol{\mu}$ where $\boldsymbol{\mu} \stackrel{\text{def}}{=} \sum_{i\in[N]} y^i \cdot \mathbf{x}^i$ and $\hat{\mathbf{w}} \stackrel{\text{def}}{=} \frac{\mathbf{w}^\top}{\|\mathbf{w}\|_2}$. The Cauchy-Schwartz inequality tells us that $\hat{\mathbf{w}}^\top \boldsymbol{\mu} \leq \|\hat{\mathbf{w}}\|_2 \|\boldsymbol{\mu}\|_2 = \|\boldsymbol{\mu}\|_2$ since $\|\hat{\mathbf{w}}\|_2 = 1$ by design and the maximum is achieved when $\hat{\mathbf{w}} \parallel \boldsymbol{\mu}$ i.e. $\hat{\mathbf{w}} = c \cdot \boldsymbol{\mu}$ for some $c \in \mathbb{R}$. Thus, the maximum is achieved at $\mathbf{w} = c \cdot \boldsymbol{\mu}$ and the desired unit norm solution is $\hat{\mathbf{w}} = \frac{\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2}$.

Note: derivations are not needed for b1 and b2. They are given here for sake of clarity. To solve (**b1**), note that for Melba's dataset, symmetry gives $\boldsymbol{\mu} = 2\left(\left(\frac{N}{2} - 1\right)\begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}1\\-1\end{pmatrix}\right) = \begin{pmatrix}N\\N-4\end{pmatrix}$ i.e. $\hat{\mathbf{w}} = \frac{1}{\sqrt{2N^2-8N+16}} \cdot \begin{pmatrix}N\\N-4\end{pmatrix} = \frac{1}{\sqrt{2-\frac{8}{N}+\frac{16}{N^2}}} \cdot \begin{pmatrix}1\\1-\frac{4}{N}\end{pmatrix}$. Using $\frac{1}{N} \approx 0$ gives $\hat{\mathbf{w}} \approx \frac{1}{\sqrt{2}} \cdot \begin{pmatrix}1\\1\end{pmatrix}$. To solve (**b2**), if the classifier is $\mathbf{w} = (a, \epsilon)$, then to classify all points correctly, we need $a + \epsilon > 0$ and $a - \epsilon > 0$ which gives $a > |\epsilon| \geq 0$. The minimum margin on any point is thus $\frac{\min\{a-\epsilon, a+\epsilon\}}{\sqrt{a^2+\epsilon^2}} = \frac{a-|\epsilon|}{\sqrt{a^2+\epsilon^2}}$. For any value of $a > 0$, the minimum margin grows if $\epsilon \to 0$. Thus, max margin is achieved at $\epsilon = 0$ and the hard SVM problem becomes $\min_a \frac{1}{2}a^2$ such that $a \geq 1$. This is achieved at $a = 1$. Thus, the hard SVM will yield the solution $\mathbf{w} = (1,0)$.

(b) Melba created a large dataset $N = 10^{101}, d = 2$ where Melbo's method fails. There are an equal number of points with $+1$ and -1 labels. One +ve point has feature $\mathbf{x}^i = (1, -1)$). All other +ve points have feature $\mathbf{x}^i = (1,1)$. One -ve point has $\mathbf{x}^i = (-1,1)$. All other -ve points have $\mathbf{x}^i = (-1,-1)$. Note that $\mathbf{x}^i \in \mathbb{R}^2, \mathbf{w} \in \mathbb{R}^2$.There is no bias term in the model (not even a hidden one).



(b1) What unit-norm classifier $\mathbf{w} \in \mathbb{R}^2$ would Melbo's method learn on Melba's dataset? (assume $10^{-100} \approx 0$)

$$\mathbf{w} = [1/\sqrt{2}, 1/\sqrt{2}]$$

(b2) What classifier $\mathbf{w} \in \mathbb{R}^2$ would hard SVM (no slacks) learn on this dataset? This vector need not be unit norm.

$$\mathbf{w} = [1, 0]$$